COMPTIA SECURITY+ SY0-701 | DOMAIN 3 · SECURITY ARCHITECTURE (18%)

Security+ 3.1 · Infrastructure Considerations

Mestre CyberSec · **Lesson 3.1 — Infrastructure Considerations** (13:48) · Assistir Vídeo no YouTube ↗ · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Considerações de design: RESILIENCY (sobreviver a falha — cluster, load balancing, RAID, multi-AZ/Região); CUSTO (CAPEX compra servidor vs OPEX pay-per-use); RESPONSIVENESS (latência — CDN perto do usuário); SCALABILITY (scale out = +máquinas é melhor).

💡 Key Exam Highlights & Critical Concepts

- 99.9% de uptime = ~8h45 por ano permitindo downtime; 99.99% = 52 min
- Elasticidade: escala pra cima E pra baixo conforme demanda
- Redundância N+1 = sobrevive 1 falha; 2N = sobrevive 2

🎯 Exam Tips & Typical CompTIA Scenarios

- 'ability to add/remove resources based on demand' = elasticidade
- 'uptime of 99.99%' = alta disponibilidade

🚨 Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.1 — Infrastructure Considerations)

Why is this advanced coverage required for SY0-701?: Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Resiliency

The ability of an infrastructure to withstand and recover from component failures while maintaining acceptable service levels, often using redundancy, clustering, and failover.

⚠ ***Exam Trap:*** *Se a questão pergunta como sobreviver a falhas sem interrupção, a resposta é resiliência. Alta disponibilidade é a meta, redundância é a técnica.*

• High Availability (HA)

A design approach that ensures a system remains operational for a very high percentage of time, commonly measured by uptime such as 99.9% (~8h45 downtime/year) or 99.99% (~52 min/year).

⚠ ***Exam Trap:*** *99,99% de uptime = ~52 min por ano. Alta disponibilidade foca em minimizar downtime, não necessariamente em sobreviver a desastres completos.*

• Scalability

The capability of a system to handle growing workloads by adding resources, either vertically (scale up) or horizontally (scale out).

⚠ ***Exam Trap:*** *'Scale out' = adicionar mais máquinas/servidores; 'scale up' = aumentar recursos de uma única máquina. CompTIA geralmente prefere scale out para resiliência.*

• Elasticity

The ability to automatically add and remove resources based on demand, scaling up and down as needed, typically in cloud environments.

⚠ ***Exam Trap:*** *Gatilho: 'ability to add/remove resources based on demand' = elasticidade. Diferente de escalabilidade, que pode ser capacidade planejada de crescer.*

• Load Balancing

Distributing incoming network traffic or workload across multiple servers to improve performance, availability, and scalability.

⚠️ ***Exam Trap:*** *Se a questão pede distribuir tráfego entre vários servidores e detectar falhas, use load balancer. Ajuda resiliência e scale out.*

• Redundancy (N+1, 2N) (N+1 / 2N)

Providing extra components beyond the minimum required; N+1 survives one failure, while 2N provides a full duplicate and can survive two failures or a complete set failure.

⚠️ ***Exam Trap:*** *N+1 = sobrevive a 1 falha; 2N = sobrevive a 2 falhas. Não confunda com níveis de RAID.*

• Capital Expenditure / Operational Expenditure (CAPEX/OPEX)

CAPEX is upfront purchase of infrastructure (e.g., servers), while OPEX is ongoing pay-per-use or subscription cost (e.g., cloud services).

⚠️ ***Exam Trap:*** *Nuvem geralmente converte CAPEX em OPEX. Se a questão fala de custo inicial vs custo recorrente, pense CAPEX vs OPEX.*

• Content Delivery Network (CDN)

A geographically distributed network of proxy servers that caches content closer to users to reduce latency and improve responsiveness.

⚠️ ***Exam Trap:*** *Gatilho: latência baixa para usuários globais = CDN. Responsividade é sobre tempo de resposta, não throughput.*

🏛️ Conceptual Deep Dive, Real-World Analogies & GRC Analysis

📌 High Availability (HA): Redundancy and Eliminating Single Points of Failure (SPOF)

🏠 **Real-World Analogy:** Think of commercial jet airliner engineering: it doesn't fly with a single engine or solitary brake cable; it features dual engines and three parallel hydraulic circuits. If primary hydraulic pressure fails mid-flight, secondary lines engage seamlessly without passengers or pilots ever losing control.

🔍 **Root Cause & Background:** Hardware components inevitably degrade and fail over time. High Availability architectures eradicate Single Points of Failure (SPOFs) by duplicating power supplies, NICs, switches, and clusters in active-active or active-passive topologies with automated health-check failover.

🛡️ **GRC Auditor Perspective:** Conduct scheduled automated failover simulations and chaos drills; audit Service Level Agreements (SLAs) to verify whether multi-zone redundancy satisfies contractually guaranteed 'five nines' (99.999%) uptime commitments.

⚠️ **How CompTIA Tests This:** Trigger: 'system designed to eliminate single points of failure and ensure continuous operational uptime during hardware crashes'. Answer: High Availability (HA).

📌 Scalability vs. Elasticity (Horizontal, Vertical, and Dynamic Auto-scaling)

🏠 **Real-World Analogy:** Scalability is expanding operational capacity: swapping a compact car for an 18-wheel cargo hauler is Vertical Scaling (Scale-Up); dispatching 10 identical compact cars is Horizontal Scaling (Scale-Out). Elasticity is a rubber band: during a 4-hour flash sale spike, the system automatically stretches to provision 50 additional cloud servers; when traffic subsides at midnight, it snaps back down to zero extra servers to eliminate waste.

🔍 **Root Cause & Background:** Long-term data growth demands structural capacity upgrades (scalability). Conversely, volatile intraday traffic demands automated, event-driven resource allocation and decommissioning (elasticity) governed by auto-scaling threshold caps.

🛡️ **GRC Auditor Perspective:** Verify that auto-scaling group policies mandate hard ceilings (maximum instance caps) and real-time billing anomaly webhooks to defend against predatory or runaway resource consumption.

⚠️ **How CompTIA Tests This:** Trigger: 'system dynamically provisions and de-provisions resources automatically in real-time as demand increases and decreases'. Answer: Elasticity. If 'adding more CPU/RAM to a single virtual server' -> Vertical Scaling (Scale-up).

📌 Latency and Geographic Proximity: CDNs and Edge Caching

🏠 **Real-World Analogy:** Imagine a bookstore housing its sole printing press in Tokyo: whenever a customer in London orders a novel, it takes 3 weeks to ship via freighter (high latency). To solve this, the publisher stocks regional newsstands throughout London with the top 50 bestsellers: customers walk 2 minutes to retrieve their book immediately. That is a CDN (Content Delivery Network).

🔍 **Root Cause & Background:** Geographical packet travel across global transoceanic backbones introduces physical latency and packet jitter. CDNs cache web assets at distributed edge Points of Presence (PoPs) while serving as defensive shock absorbers against volumetric DDoS attacks.

🛡️ **GRC Auditor Perspective:** Inspect TLS termination configurations on CDN edge endpoints; verify cache-control directives ('Cache-Control: private, no-store') to prevent confidential dynamic PII payloads from caching in edge nodes.

⚠️ **How CompTIA Tests This:** Trigger: 'distributed network of edge proxy servers used to reduce latency for global users and absorb distributed denial-of-service (DDoS) floods'. Answer: CDN (Content Delivery Network).

📌 Disaster Recovery (DR): Regions, Availability Zones, RPO, and RTO

🏠 **Real-World Analogy:** An Availability Zone (AZ) is operating two separate office buildings across town on distinct municipal power grids: if one flooded, the other runs uninterrupted. A Cloud Region is maintaining a second facility across the ocean in a different country: immune to local earthquakes or regional grid blackouts. RPO (Recovery Point Objective) is how many hours of receipts you can afford to lose forever (data loss tolerance); RTO (Recovery Time Objective) is how long before the emergency backup doors swing open to customers (downtime tolerance).

🔍 **Root Cause & Background:** Natural disasters, fiber cuts, and regional power-grid blackouts incapacitate entire physical facilities. Multi-AZ and multi-region replication architectures ensure business survival against catastrophic localized failures.

🛡️ **GRC Auditor Perspective:** Audit the enterprise Business Impact Analysis (BIA); benchmark disaster recovery failover drill metrics to validate that empirical replication timings conform strictly to board-mandated RPO and RTO targets.

⚠️ **How CompTIA Tests This:** Trigger: 'maximum acceptable amount of data loss measured in time between last backup and disaster event'. Answer: RPO (Recovery Point Objective). If 'maximum acceptable duration of time to restore system operations' -> RTO (Recovery Time Objective).

⚡ QUICK-REFERENCE EXAM CHEAT SHEET

Typical Exam Scenario / Practical Situation	Technical Concept & Target Answer
Architecture engineered with redundant hardware and automated failover to ensure operational uptime without SPOFs	**High Availability (HA)**
Dynamic expansion and de-provisioning of cloud server capacity in real-time response to immediate workload fluctuations	**Elasticity**
Adding additional CPU sockets and memory modules to upgrade the physical hardware capacity of an existing machine	**Vertical Scaling (Scale-up)**
Geographically dispersed caching infrastructure caching assets near end users to slash latency and mitigate DDoS floods	**CDN (Content Delivery Network)**
Maximum tolerable duration of operational downtime permissible before business processes are brought back online	**RTO (Recovery Time Objective)**
Maximum permissible parameter of irreversible data loss measured in time elapsed between the last backup and outage	**RPO (Recovery Point Objective)**

✅ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A retail company runs its online store on a single physical web server in one data center. Last month, a failed power supply took the store offline for four hours. Management wants the store to remain available if any single web server fails, without changing the application architecture. Which infrastructure design should the security administrator recommend?

A) Deploy two or more web servers behind a load balancer in separate availability zones.

B) Replace the server with a larger model that has redundant power supplies and more memory.

C) Configure RAID 10 on the existing server and increase backup frequency.

D) Move the web server to a VLAN with redundant switches and firewall rules.

QUESTION 2 · EXAM SCENARIO

A startup's finance team wants to avoid large upfront purchases of servers and storage. They want infrastructure costs to vary with monthly customer usage and be treated as an operating expense. Which cloud purchasing model best meets this requirement?

A) Reserved instances paid in full for three years

B) On-premises servers purchased through the capital budget

C) Infrastructure as a Service with pay-as-you-go consumption billing

D) A colocation contract with owned hardware and fixed power fees

QUESTION 3 · EXAM SCENARIO

A media company hosts videos and images for users across North America, Europe, and Asia. Users far from the origin data center report slow page loads because static content takes a long time to download. Which solution most directly improves responsiveness for these users?

A) Deploy a content delivery network to cache content at edge locations.

B) Add more RAM to the origin web servers.

C) Configure database replication within the origin region.

D) Implement RAID 6 on the storage array that holds the media files.

QUESTION 4 · EXAM SCENARIO

An online ticketing platform experiences a sharp increase in traffic when tickets are released. The current web server reaches 100% CPU during each release. Administrators want to improve capacity by using multiple smaller web servers instead of moving to a larger, more expensive server. Which approach should they implement?

A) Scale out by adding web server instances behind a load balancer.

B) Scale up by increasing vCPUs and memory on the existing server.

C) Implement RAID 10 on the web server's local disks.

D) Deploy a CDN to cache ticket inventory API responses.

QUESTION 5 · EXAM SCENARIO

A SaaS provider sees user demand rise every weekday morning and fall at night. The security architect wants the application tier to automatically add virtual machines during peak hours and remove them during low usage to match demand while controlling cost. Which concept should be enabled?

A) Elasticity through auto-scaling

B) Resiliency through RAID

C) Responsiveness through CDN

D) Redundancy through 2N power

QUESTION 6 · EXAM SCENARIO

A data center has four cooling units that must all operate to maintain safe temperature during peak summer load. After a recent failure, management asks the security administrator to add capacity so that one cooling unit can fail without affecting operations. Which redundancy model should be implemented with the minimum extra capacity?

A) N

B) N+1

C) 2N

D) N+2

QUESTION 7 · EXAM SCENARIO

A financial services firm hosts a critical trading application in a single cloud region with three availability zones. The business continuity plan now requires the application to remain operational even if an entire cloud region becomes unavailable. Which design change best meets this requirement?

A) Deploy the application across multiple cloud regions with active-active failover.

B) Add a fourth availability zone inside the existing region.

C) Replace local disks with a RAID 10 array on each database server.

D) Configure a global load balancer to distribute traffic only within the current region.

QUESTION 8 · EXAM SCENARIO

A company uses a load balancer in front of two web servers. During testing, one web server's application process hangs, but the server still responds to ICMP ping. The load balancer continues to send user requests to the hung server, causing failed transactions. Which configuration change should the administrator make?

A) Enable HTTP/HTTPS health checks that test a specific application endpoint.

B) Increase the load balancer's TCP timeout to several minutes.

C) Enable sticky sessions so users remain on the same server.

D) Add a second load balancer in an active-passive pair.

QUESTION 9 · EXAM SCENARIO

A database server requires high disk read/write performance and must continue operating if a single drive fails. The administrator is selecting a RAID configuration. Which RAID level best meets both requirements?

A) RAID 0

B) RAID 1

C) RAID 5

D) RAID 10

QUESTION 10 · EXAM SCENARIO

A manufacturing company has a stable, predictable workload and strict data residency rules. It already owns a data center with unused capacity and wants to minimize long-term infrastructure cost. Which infrastructure strategy is most appropriate?

A) Migrate all workloads to a public cloud with pay-per-use billing.

B) Use a CDN to cache internal manufacturing databases.

C) Implement auto-scaling across multiple public cloud regions.

D) Continue using the existing on-premises data center with owned hardware.

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — Deploy two or more web servers behind a load balancer in separate availability zones.
A load balancer with multiple web servers in separate availability zones provides redundancy and failover if one server fails. Dual power supplies protect only the power component, RAID protects disks, and redundant network devices do not address a failed server.

Question 2: Option C — Infrastructure as a Service with pay-as-you-go consumption billing
IaaS with pay-as-you-go consumption billing is an operational expense because the organization pays only for what it uses without large upfront hardware purchases. Reserved instances involve a commitment and possible upfront payment, while on-premises and colocation with owned hardware are capital expenses.

Question 3: Option A — Deploy a content delivery network to cache content at edge locations.
A content delivery network caches static content at edge locations closer to users, reducing latency and improving responsiveness. Adding RAM, replicating databases regionally, or using RAID does not reduce the network distance to global users.

Question 4: Option A — Scale out by adding web server instances behind a load balancer.
Scaling out adds more web server instances and distributes load across them, increasing capacity horizontally. Scaling up increases resources on one server, RAID addresses disk redundancy, and a CDN is intended for cacheable content rather than scaling application processing.

Question 5: Option A — Elasticity through auto-scaling
Elasticity through auto-scaling automatically adjusts the number of resources based on demand. RAID, CDN, and 2N power redundancy address disk failures, latency, and power failures, respectively, not dynamic capacity matching.

Question 6: Option B — N+1
N+1 provides one additional component beyond the required N, allowing the failure of one cooling unit. 2N and N+2 provide more redundancy than the minimum needed, while N provides no tolerance for failure.

Question 7: Option A — Deploy the application across multiple cloud regions with active-active failover.
To survive the loss of an entire cloud region, the application must be deployed across multiple regions with failover. Adding another availability zone in the same region, using RAID locally, or balancing only within the current region does not protect against a regional outage.

Question 8: Option A — Enable HTTP/HTTPS health checks that test a specific application endpoint.
Application-layer health checks test whether the web application is actually responding, detecting a hung process even when the host replies to ping. Increasing timeout, enabling sticky sessions, or adding a redundant load balancer does not detect the failed application process.

Question 9: Option D — RAID 10
RAID 10 combines mirroring and striping, providing high performance and tolerance for a single drive failure. RAID 0 has no redundancy, RAID 1 may have lower write performance, and RAID 5 can have slower writes and rebuilds.

Question 10: Option D — Continue using the existing on-premises data center with owned hardware.
For a predictable workload, strict data residency needs, and existing owned capacity, continuing with on-premises infrastructure can minimize long-term cost. Public cloud pay-per-use, CDN for databases, and multi-region auto-scaling can add cost and complexity without matching the stated requirements.